

Network Infrastructure Concepts

Wired Networks



Network Infrastructure Concepts

- Wired Networks
- Network Hardware Devices
- Wireless Networks
- Internet Connection Types
- Network Configuration Concepts
- Network Services

Network Types (Slide 1 of 4)

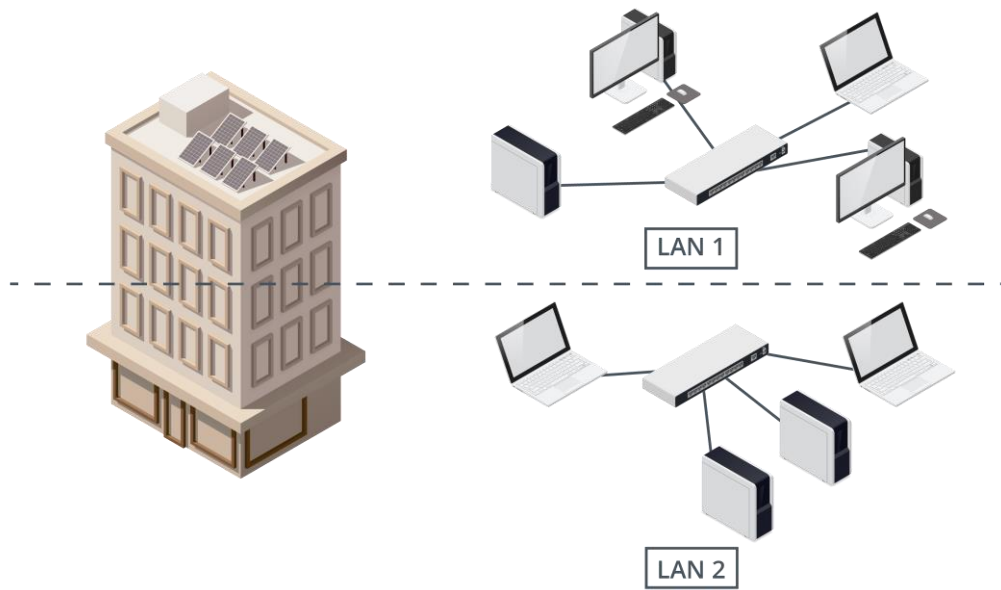


Network: In its most simple form, a network consists of two or more computers connected to each other by an appropriate transmission medium which allows them to share data.

- Purpose: provide services and resources to users
- Historically: files, folders, printers, email, databases
- Modern: web applications, social networking, VoIP, multimedia conferencing
- Types: LANs, WANs, MANs

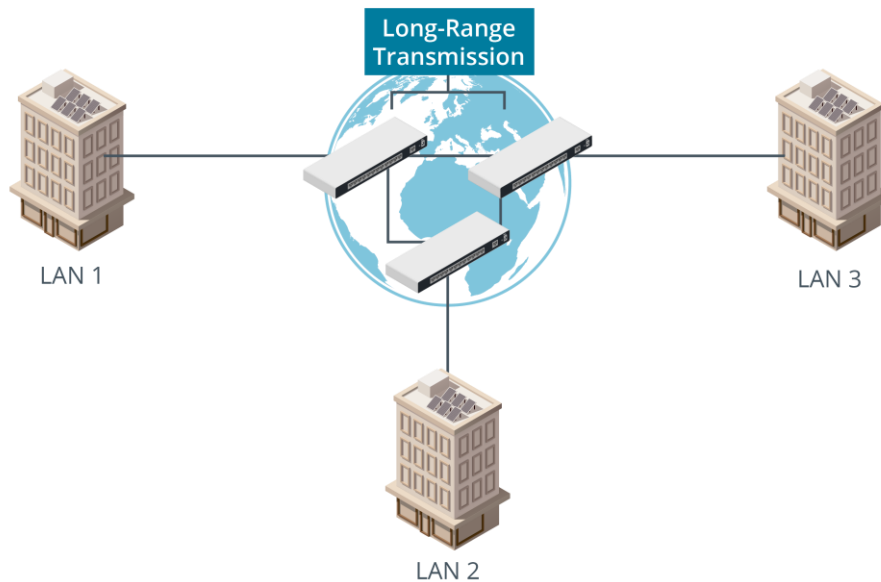
Network Types (Slide 2 of 4)

- LANs within a building



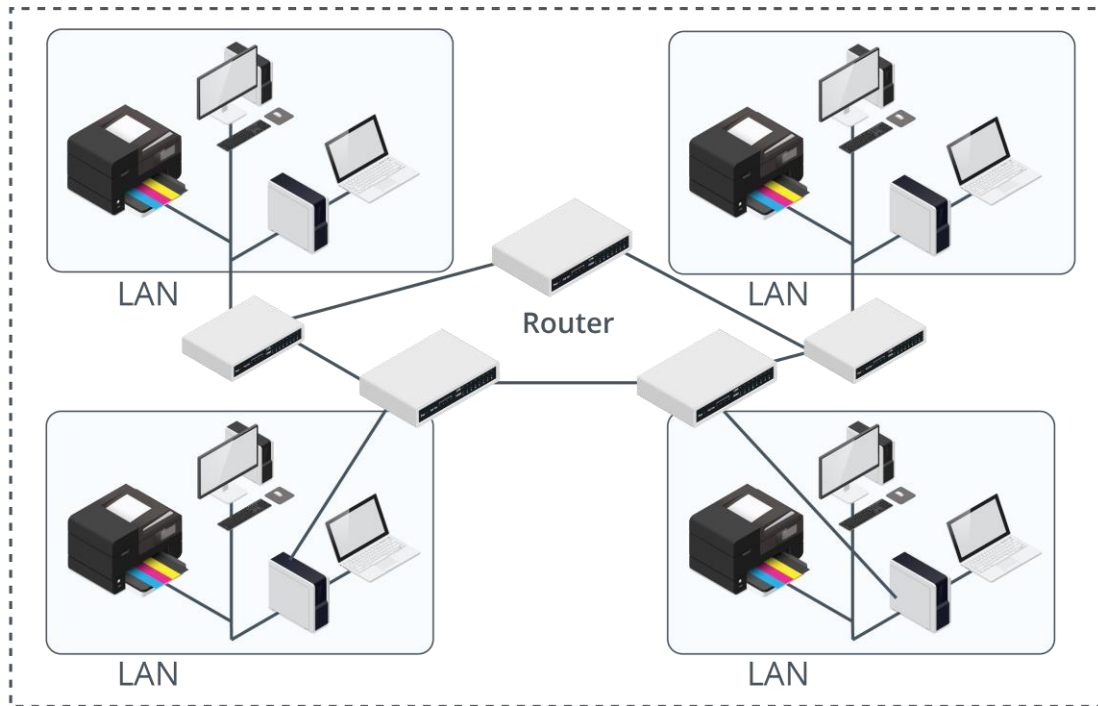
Network Types (Slide 3 of 4)

- A Wide Area Network (WAN)



Network Types (Slide 4 of 4)

- A Metropolitan Area Network (MAN)

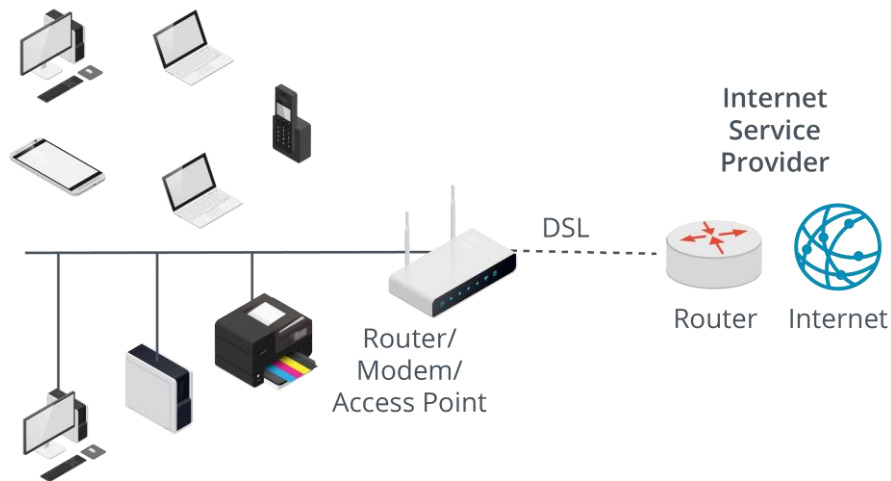


Ethernet Types and Standards

- Most cabled LANs build on Ethernet
 - Developed by DIX consortium
 - Maintained by IEEE (802.3 standards)
- Types:
 - 10 Mbps (10BASE)
 - Fast Ethernet (100BASE)
 - Gigabit Ethernet (1000BASE)
 - 10G Ethernet (10GBASE)
- IEEE 802.11 standards (Wi-Fi) for WLANs are complementary
- Flexible, self-contained, scalable

Common Ethernet Network Implementations (Slide 1 of 3)

- SOHO: business network with server and clients, using single Internet appliance as access point, Ethernet switch, Internet modem, Internet router.

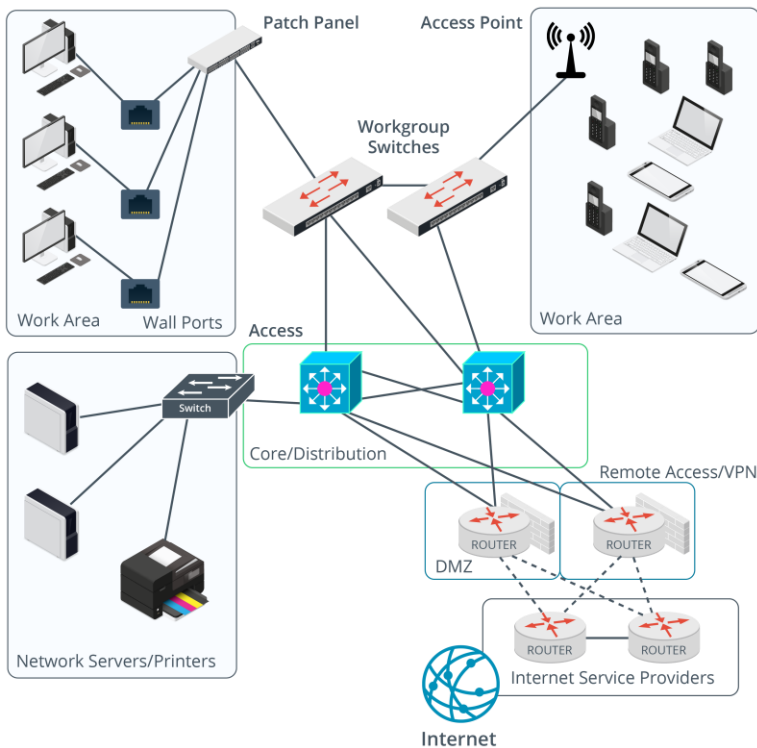


Common Ethernet Network Implementations (Slide 2 of 3)

- Enterprise network: usually dedicated single-function connectivity appliances.
 - Small and Medium (SME): tens of users; multiple switches, access points, routers.
 - Enterprise LAN: hundreds or thousands of servers and clients; multiple enterprise-class switches, access points, and routers.
 - Campus Area Network (CAN) = LAN spanning multiple nearby buildings.

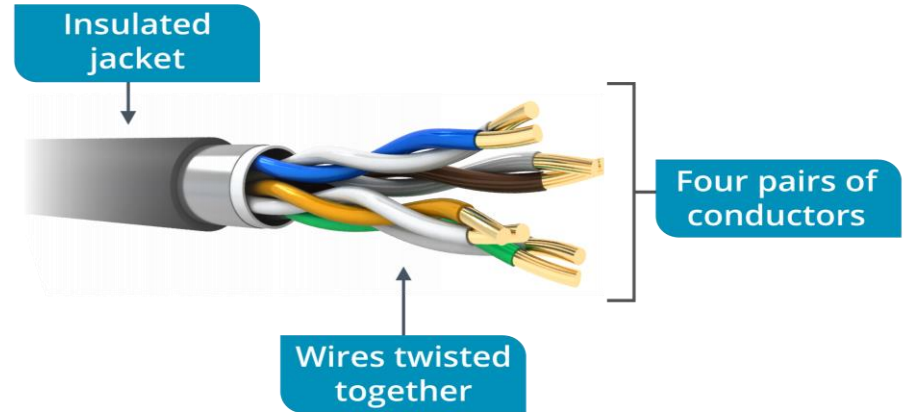
Common Ethernet Network Implementations (Slide 3 of 3)

- Positioning network components in an enterprise LAN



Twisted Pair Cabling and Connectors (Slide 1 of 4)

- Unshielded Twisted Pair (UTP)
 - Most widely used
 - Four copper conductor pairs
 - Insulating sheath
 - Twisted to reduce crosstalk and EMI
 - Paired wires carry equal/opposite signals
 - PVC jacket
 - Works well in low interference; has limited range, may exhibit attenuation



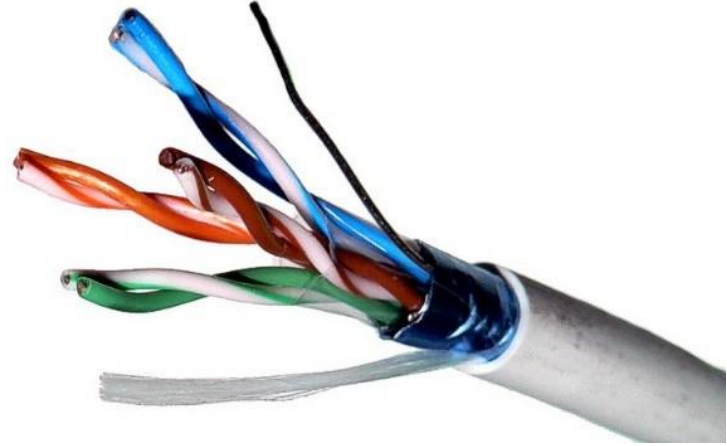
Twisted Pair Cabling and Connectors (Slide 2 of 4)

- Cat standards:

Cat	Frequency	Capacity	Max. Distance	Network Applications
5	100 MHz	100 Mbps	100 m (328 ft)	100BASE-TX
5e	100 MHz	1 Gbps	100 m (328 ft)	1000BASE-T
6	250 MHz	1 Gbps	100 m (328 ft)	1000BASE-T
6	250 MHz	10 Gbps	50 m (180 ft)	10GBASE-T
6A	500 MHz	10 Gbps	100 m (328 ft)	10GBASE-T

Twisted Pair Cabling and Connectors (Slide 3 of 4)

- Shielded Twisted Pair (STP)
 - Originally used braided shield to reduce interference and crosstalk; can be bulky, difficult to install.
 - Modern STP uses screened cables; shield positioned around all pairs.
 - Shielded Cat 53/6/6A:
 - F/UTP (also ScTP)
 - U/FTP
 - Modern STP solutions incorporate grounding in each element.



Twisted Pair Cabling and Connectors (Slide 4 of 4)

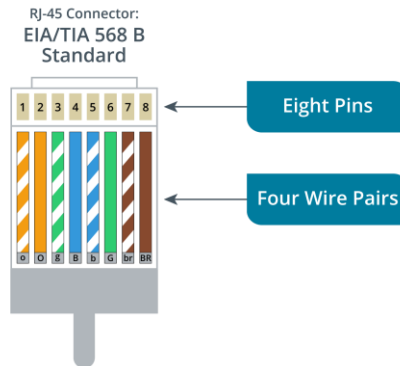
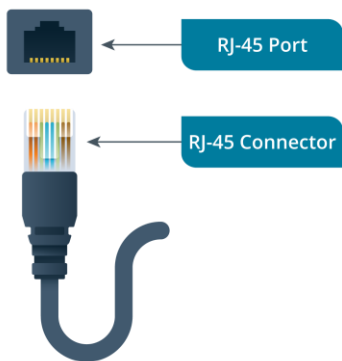


Plenum: An air handling space, including ducts and other parts of the HVAC system in a building.

- Plenum space:
 - Typically a false ceiling, may be raised floor.
 - May be used for communications wiring.
 - Can be conduit for fire.
- General purpose non-plenum cable uses PVC; marked CMG/MMG or CM/MP
- Plenum cable:
 - Must not emit smoke, must self-extinguish, meet other fire safety standards.
 - Uses treated PVC or FEP; can be less flexible, does not affect bandwidth.
 - Marked CMP/MMP.

Wiring Standards for Twisted Pair (Slide 1 of 2)

- Ethernet twisted pair terminated with RJ-45 connectors:
 - 8P8C (8-position/8-contact)
 - Color-coded (Blue, Orange, Green, Brown)
 - 1st conductor in pair has white/stripes
 - 2nd conductor in pair is solid color
- ANSI/TIA/EIA 568 termination:
 - T568A is shown
 - T568B:
 - Pin 1=Orange/White; Pin 2=Orange; Pin 3=Green/White; Pin 4=Green



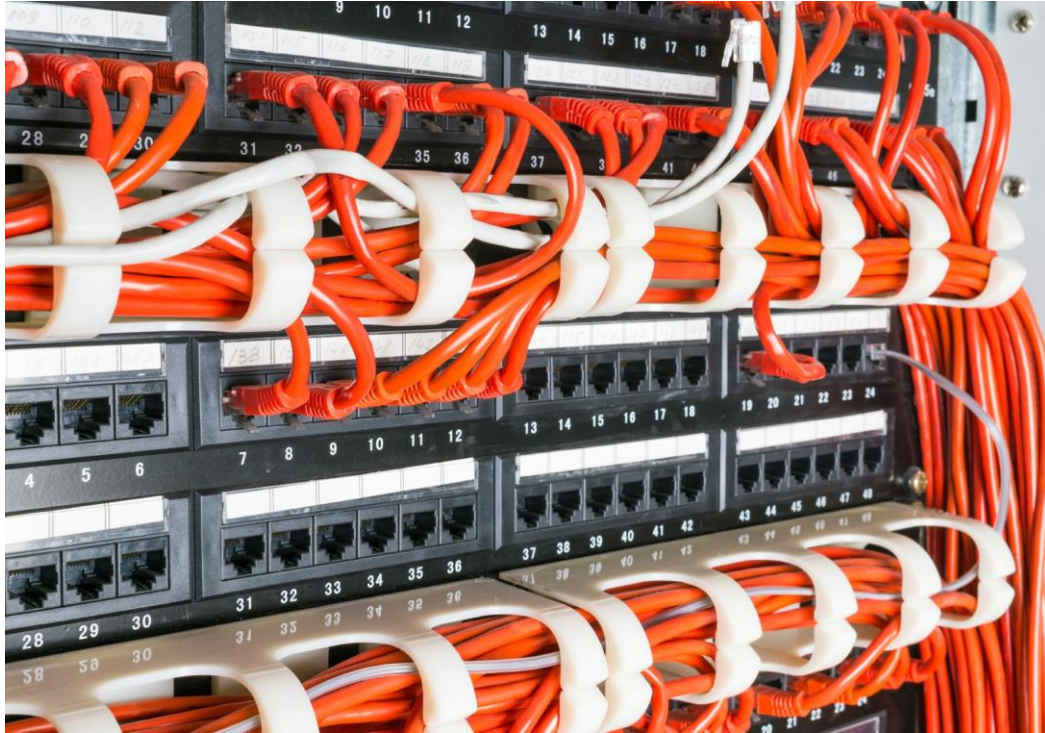
Wiring Standards for Twisted Pair (Slide 2 of 2)

- Normal (straight-through) cable has same termination at both ends.
- Crossover cable has T568A at one end, T568B at other.
 - Previously used for direct connections.
 - Now Gigabit Ethernet interfaces can automatically cross over with standard cable.
- Avoid mixing standards:
 - Both are common.
 - T568A mandated for US government and by TIA 570 residential cabling standard.

Patch Panels and Structured Cabling (Slide 1 of 2)

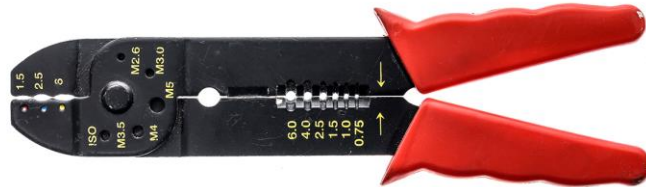
- Gigabit Ethernet: no more than 100 m of cable between switch and computer
- Solid cabling:
 - Single thick wire for permanent links, aka “drop cables”
 - Links RJ-45 port on wall plate with patch panel
 - Terminates in IDC
- Patch cord connects RJ-45 port on panel to port on switch
 - Stranded cable; flexible, less efficient
 - 5 m maximum length
- Second patch cord from computer to wall
- Structured cabling system

Patch Panels and Structured Cabling (Slide 2 of 2)



Cable Installation and Testing Tools (Slide 1 of 7)

- Wire stripper/cutter: for cutting wire and stripping insulation and cable jackets.



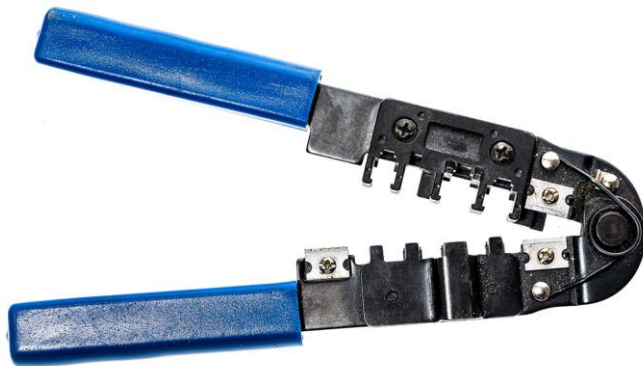
Cable Installation and Testing Tools (Slide 2 of 7)

- Punch-down tool: fixes conductors into an IDC.



Cable Installation and Testing Tools (Slide 3 of 7)

- Crimpers: fix a jack into a cable.



Cable Installation and Testing Tools (Slide 4 of 7)

- Cable testing:
 - Verify wiring installation and termination just after making connections, with access to cable runs.
 - Simpler than during user device setup.
 - Consider:
 - Patch cord between PC and wall.
 - Wall port and wall cabling.
 - Port on patch panel and patch cord to switch port.
 - Test with a known good cable.
 - Various troubleshooting devices.

Cable Installation and Testing Tools (Slide 5 of 7)

- Multimeter: basic cable testing tool; tests for copper wire continuity, existence of short, integrity of terminator.
- Wire map tester: identifies transpositions and reverse pairs.
- Advanced testers: show cable's physical/electrical properties.
- Certifiers: test and certify installation to a category.



Multimeter



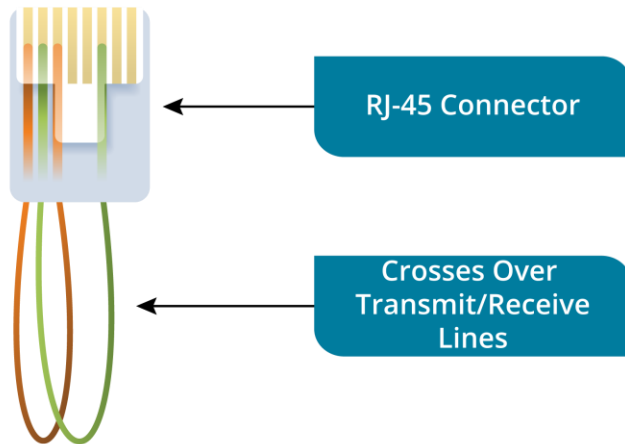
Cable tester

Cable Installation and Testing Tools (Slide 6 of 7)

- Tone generator and probe (aka “fox and hound” ; “tone and probe”): traces cable from end to end.
- Connect generator to wires, move locator over cable group until beep is loudest.

Cable Installation and Testing Tools (Slide 7 of 7)

- Loopback plug: tests a port
- Connects pin 1 to pin 3 and pin 2 to pin 6



Fiber Optic Cabling and Connectors (Slide 1 of 3)

- Electrical signals on copper wire subject to interference/attenuation.
- Light signals on fiber optic cable resist interference, eavesdropping, attenuation.
- Supports higher bandwidth, longer cable runs.

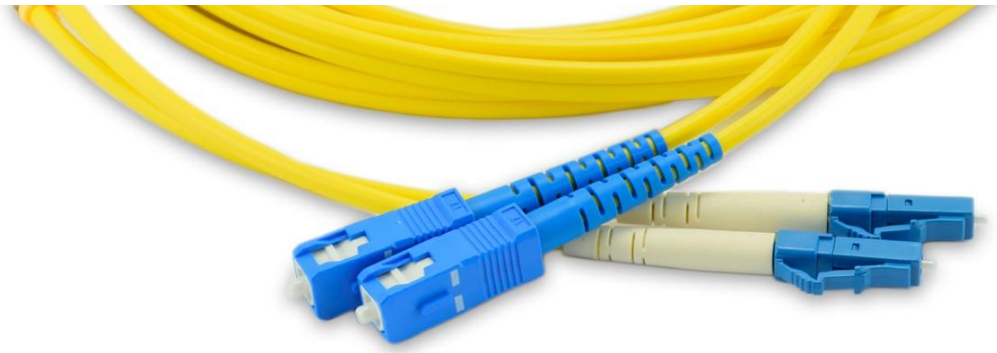


Fiber Optic Cabling and Connectors (Slide 2 of 3)

- Single-Mode Fiber (SMF)
 - Small core, long wavelength, near-infrared signal generated by laser.
 - Data rates up to 10 Gbps or more; cable runs of many kilometers (suitable for WANs).
- Multi-Mode Fiber (MMF)
 - Larger core, shorter wavelength.
 - Less expensive optics, less expensive deployment.
 - Lower signaling speeds, shorter distances (suitable for LANs).

Fiber Optic Cabling and Connectors (Slide 3 of 3)

- Connector types:
 - Straight Tip (ST), Subscriber Connector (SC), Lucent/Local Connector (LC).
 - Patch cords can have same or mixed connectors.
 - Connectors damage easily; plug/unplug only when needed.



Coax Cabling and Connectors (Slide 1 of 4)

- Two conductors share the same axis.
- Signal conductor insulated; second wire mesh conductor acts as EMI shield and as ground.



Coax Cabling and Connectors (Slide 2 of 4)

- Radio Grade (RG) “standard”; developed by US military, categorizes cable by thickness and impedance.
 - RG-6: thicker core, better quality, often used as drop/patch cable in modern CATV and broadband.
 - RG-59: thinner core; drop cable for older CATV/cable modems; used in CCTV.
- Coax also available with tri- or quad-shielding.

Coax Cabling and Connectors (Slide 3 of 4)

- BNC connectors at cable ends in most cases.
- BNC couplers can connect cables .
- Impedance of connector must match cable type (50 or 75 ohm).
- Also screw-down F-connectors.

Coaxial
F-Connector



BNC
Connector



Coax Cabling and Connectors (Slide 4 of 4)

- 10BASE-5/Thicknet and 10BASE-2/Thinnet supported 10 Mbps, up to 500 m and 185 m, respectively.
- Coax now obsolete for LANs; in use for CCTV and drop cables for CATV and Internet.
- Hybrid Fiber Coax (HFC): Coax links fiber trunk in street to customer cable modem.
- Less attenuation than TP but bulkier, harder to install.

Activity



Discussing Wired Networks

Network Infrastructure Concepts

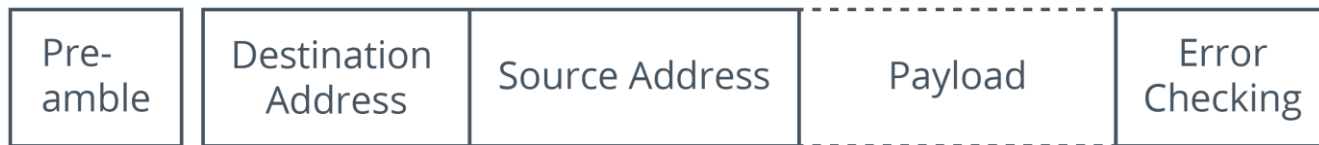
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Network Interface Cards (Slide 1 of 2)

- Network Interface Card (NIC) port provides connection to network media.
- Data signals must come in regular units with consistent format.
- Each node must be able to address other nodes.
- Ethernet data link protocol provides addressing, framing functions.
- Various encoding mechanisms; NIC transceiver transmits and sends in agreed frame format.

Network Interface Cards (Slide 2 of 2)

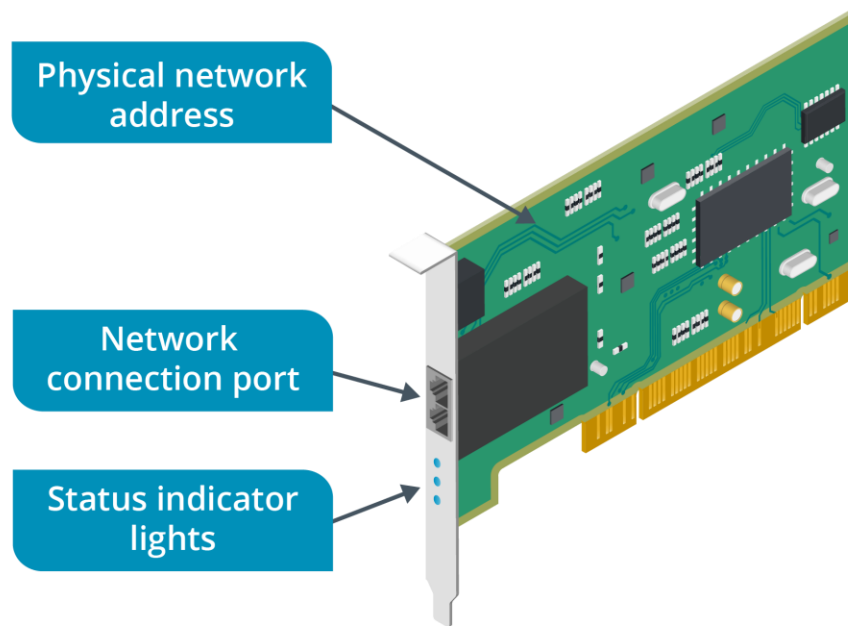
- Construction of a frame:



Ethernet NIC Features (Slide 1 of 4)

- Usually on board the motherboard.
- May be PCIe expansion board.
- All onboard cards support copper-based Ethernet with RJ-45 ports.
- Expansion cards may support:
 - Fiber optic.
 - Multiple port types.
 - Multiple ports of same type (can be bonded for higher-speed link).

Ethernet NIC Features (Slide 2 of 4)



Ethernet NIC Features (Slide 3 of 4)

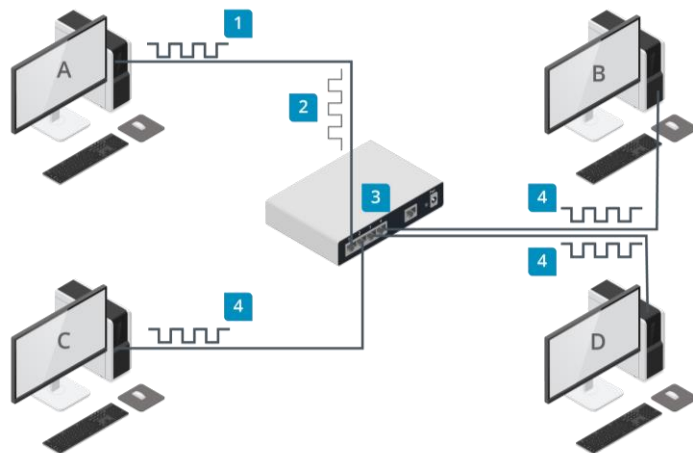
- MAC address: unique address for each Ethernet adapter port.
- Provides value for frame source and destination fields.
- 48 bits/6 bytes.
- Shown as 12 hex digits:
 - May have colon, hyphen, or no separator.
 - Examples: 00:60:8c:12:3a:bc or 00608c123abc.

Ethernet NIC Features (Slide 4 of 4)

- LED status lists show connection status:
 - Link light shows if network signal present.
 - Activity light flickers when packets received/sent.
 - Speed light possible on multi-speed adapters.
 - Dual-color LEDs combine functions.

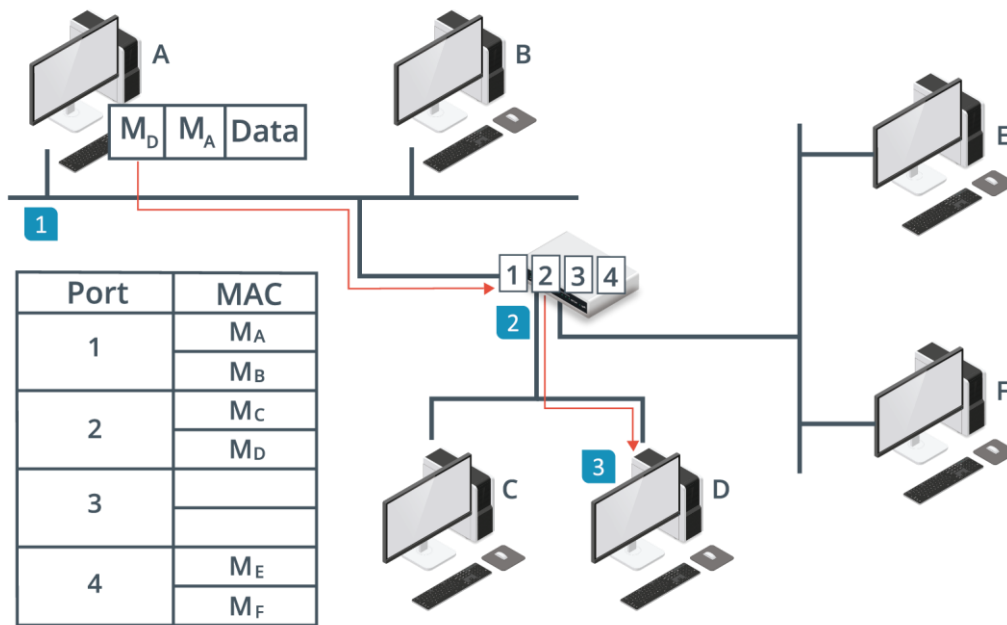
Legacy Networking Devices (Slide 1 of 2)

- Switch is appliance at core of modern networks.
- Legacy appliances include:
 - Hub: center of Ethernet star topology, works as multiport repeater.
 - Repeater: retransmits signal to overcome distance limitations.
 - Bridge: divides network into segments (collision domains) to reduce contention and collision.



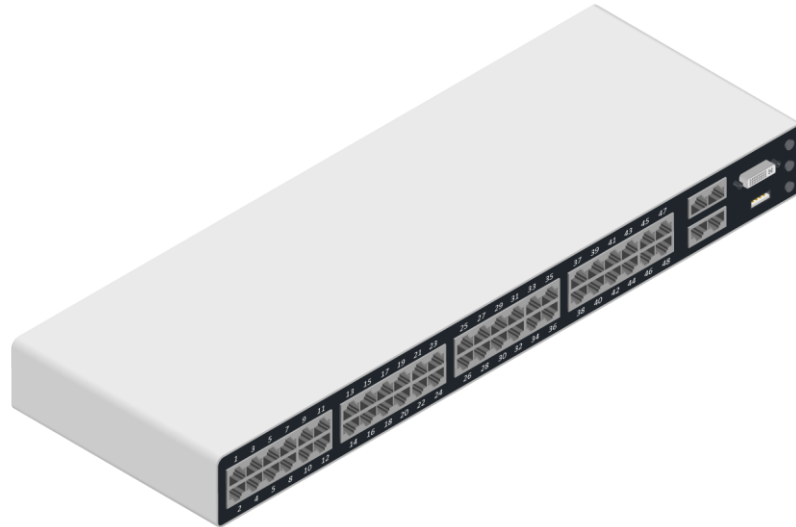
Legacy Networking Devices (Slide 2 of 2)

- Bridge operation:



Switches (Slide 1 of 3)

- Ethernet switch in modern network acts like hub, repeater, and bridge.

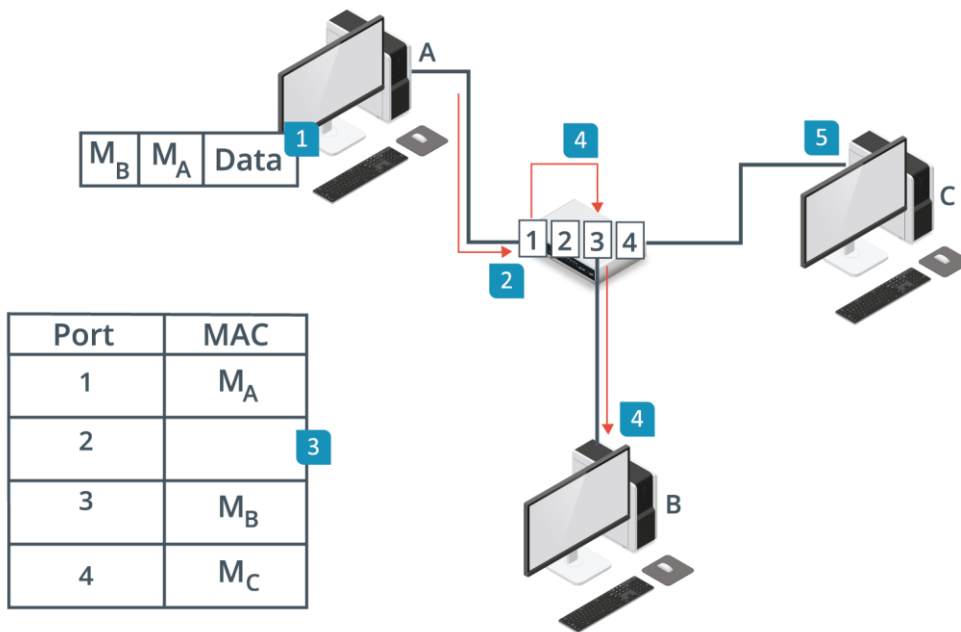


Switches (Slide 2 of 3)

- Microsegmentation:
 - Switches have up to 48 ports.
 - Multiple switches can connect into switched fabric with thousands of ports.
 - Each port is separate collision domain.
 - Establishes point-to-point link (virtual circuit) between any two nodes.
 - Collisions only occur if port is half-duplex (attached to a legacy card or node).
 - Collisions only affect that segment, not the whole network.

Switches (Slide 3 of 3)

- Switch operation:



Managed and Unmanaged Switches (Slide 1 of 3)

- Unmanaged switch:
 - Performs microsegmentation without configuration.
 - May be found in small networks (4 or 8 port switches).
 - Embedded in most ISP's Internet routers/modems.
- Managed switch:
 - For larger workgroups and corporate networks.
 - Unmanaged out of the box, but can be configured administratively.
 - Can provide thousands of access ports by linking switches.
 - Can divide into virtual LANs (VLANs).

Managed and Unmanaged Switches (Slide 2 of 3)

- Cisco Catalyst 9400 Series modular chassis



Managed and Unmanaged Switches (Slide 3 of 3)

- Interface configuration on a Cisco switch

```
FastEthernet1/0/1 is up, line protocol is up (connected)
  Hardware is Fast Ethernet, address is f41f.c253.7103 (bia f41f.c253.7103)
  MTU 1500 bytes, BW 100000 Kbit/sec, DLY 100 usec,
    reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full-duplex, 100Mb/s, media type is 10/100BaseTX
  input flow-control is off, output flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:51, output 00:00:00, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
    18 packets input, 1758 bytes, 0 no buffer
    Received 4 broadcasts (2 multicasts)
    0 runts, 0 giants, 0 throttles
    0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
    0 watchdog, 2 multicast, 0 pause input
    0 input packets with dribble condition detected
  111 packets output, 13828 bytes, 0 underruns
  0 output errors, 0 collisions, 1 interface resets
  0 unknown protocol drops
```


Power Over Ethernet

- Supplies power from a switch port over Cat 5 or better to a powered device.
- Two IEEE standards (both now in 802.3-2012):
 - 802.3af
 - 802.3at (PoE+)
- PoE-enabled switches called end-span/end-point PSE.
- Power injector can be used if switch does not support PoE.
- Switch detects if connected device is PoE-enabled.
- More efficient than powering each device through a wall socket.
- Network-management software can control devices, apply power schemes.

Ethernet Over Power (Slide 1 of 2)

- SOHO networks are unstructured, use a single router, incorporate smart appliances.
- Wireless is obvious solution; WLAN bandwidth may be adequate.
- There may be interference issues or appliances may not support Wi-Fi.
- Ethernet over Powerline uses building power circuits; overlays carrier signal to transfer Ethernet frames.
- Adapter plugs into electrical outlet; provides RJ-45 ports.
- No configuration needed, but security can be configured.
- Standards defined by IEEE 1901, managed by HomePlug Powerline Alliance.

Ethernet Over Power (Slide 2 of 2)

- Netgear Powerline AV200 adapters



Activity



Discussing Network Hardware Devices

Network Infrastructure Concepts

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What is Wireless Networking?

- A range of connectivity products and devices.
- Personal area networking to Internet connectivity.
- Usually uses radio waves for transmission, tuned to specific frequency.



Wireless Frequencies and Channels (Slide 1 of 4)

- RF ranges from 3 KHz to 300 GHz
- Subdivided into bands (FM radio and TV are VHF band)
- Radio spectrum use regulated by governments
- Standardized by ITU
- Frequency use requires a license
- Some unregulated frequencies



Wireless Frequencies and Channels (Slide 2 of 4)

- IEEE 802.11 standards = Wi-Fi
 - (Transfer rates for optimal installation; frequencies lack penetration; may be interference; data rate may drop with distance.)

Standard	Maximum Transfer Rate	Band
802.11a (1999)	54 Mbps	5 GHz
802.11b (1999)	11 Mbps	2.4 GHz
802.11g (2003)	54 Mbps	2.4 GHz
802.11n (2009)	288.8 Mbps/stream (Single Channel) 600 Mbps/stream (Bonded Channels)	2.4/5 GHz
802.11ac (2013)	1.7 Gbps (at time of writing)	5 GHz

Wireless Frequencies and Channels (Slide 3 of 4)

- Two most common frequency bands in 802.11:
 - 2.4 GHz:
 - Longer wavelength, longer range, propagates better through solids
 - Does not support many individual channels; is often congested
 - Increased risk of interference
 - Achievable data rates less than 5 GHz
 - 5 GHz:
 - Less effective at solid surface penetration
 - Lower range than 2.4 GHz
 - More individual channels; less congestion
 - Higher data rates



Wireless Frequencies and Channels (Slide 4 of 4)

- Range:
 - 2.4 GHz: maximum indoor range ~30-45 m (100–150 ft).
 - 5 GHz: maximum indoor range up to ~30 m.
 - Absolute range less important than number of clients to support and wall/ceiling construction.
- Channels:
 - 2.4 GHz: up to 14 channels, considerable overlap, co-channel interference.
 - Special codes distinguish pattern of each node.
 - Channel can become saturated.
 - 5 GHz: 23 non-overlapping channels:
 - More WAPs in same area or access points closer together, higher client device density.

Wireless Network Standards (Slide 1 of 2)

- 802.11a/b/g considered legacy standards, limited to old equipment
 - 802.11b/g were more successful
 - Both worked at 2.4 GHz; 802.11b WLANs upgraded to 802.11G
 - 802.11a works at 5 GHz; incompatible



Wireless Network Standards (Slide 2 of 2)

Standard	Description
802.11n	• More bandwidth than legacy standards. • Multiplexes 2-4 antennas using MIMO. • Ax:B:C notation (transmit antennas, receive antennas, simultaneous streams). • Can use 2.4 GHz or 5 GHz band (preferred). • Can use channel bonding in 5 GHz band to deliver more bandwidth. • Nominal data rates 288.8 Mbps (single channel) and 600 Mbps (bonded channels).
802.11ac	• Continues development of 802.11n. • Works in 5 GHz range. • Can use 2.4GHz range for legacy standards in mixed mode. • Aims for throughput comparable to Gigabit Ethernet. • Supports channel bonding to 80 or 160 MHz channels; 8 spatial streams vs. 4; denser modulation. • Needs high-end equipment for sufficient antennas for 8 streams. • Theoretical data rate with 8 streams and 160 MHz channel bonding ~6.93 Gbps.

Access Points and Wireless Network Modes (Slide 1 of 3)

- Most Wi-Fi networks are Infrastructure mode
- Each client connects with an Access Point (AP)
- Forms Basic Service Set (BSS)
- MAC address of AP is BSSID
- Can group BSS's = ESS
- AP is bridge between wired/wireless network (Distribution System)
- AP connects to network like a host computer



Access Points and Wireless Network Modes (Slide 2 of 3)

- Access points can forward frames in a Wireless Distribution System (WDS):
 - Extends network without a cabled backbone.
 - Bridge mode and repeater mode.
 - Can be complex; can be compatibility issues with different vendors' devices.
- Range extender is simpler residential solution; can work with powerline adapter.

Access Points and Wireless Network Modes (Slide 3 of 3)

Configuration	Description
Ad-hoc and Wi-Fi Direct	• Stations configured to make direct peer-to-peer connections. • Older standards: ad-hoc mode. • Modern approach: Wi-Fi Direct.
Wireless Mesh Network (MSN)	• Part of 802.11s standard. • Nodes (Mesh Stations) can discover each other and make peer connections to form Mesh Basic Service Set. • Mesh stations perform path discovery and forwarding. • Routing protocol such as Hybrid Wireless Mesh Protocol (HWMP).
Personal Area Network (PAN)	• Using wireless connectivity to connect to devices with a few meters. • Printers, smartphones, headsets, etc.

Wireless Network Cards

- Each Wi-Fi network station needs network adapter compatible with 802.11 standard in use on network.
- Onboard adapters; expansion boards; USB-connected adapters.
- Link-layer MAC address like Ethernet cards.

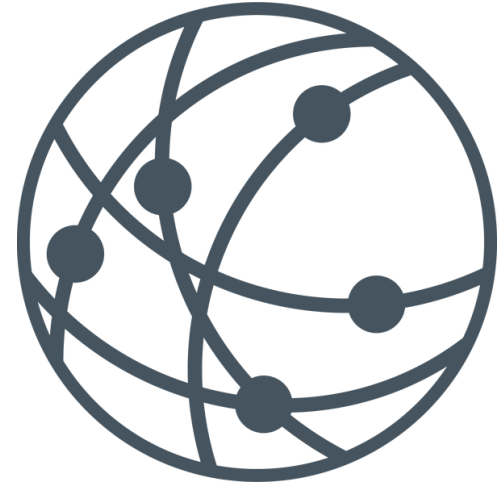
Activity



Discussing Wireless Networks

Internet Connections

- Businesses and homes depend on Internet access.
- Internet backbone: high-bandwidth backbones connecting Internet eXchange Points (IXPs).
- Created by telecommunications companies and academic institutions .
 - Organized nationally and internationally.



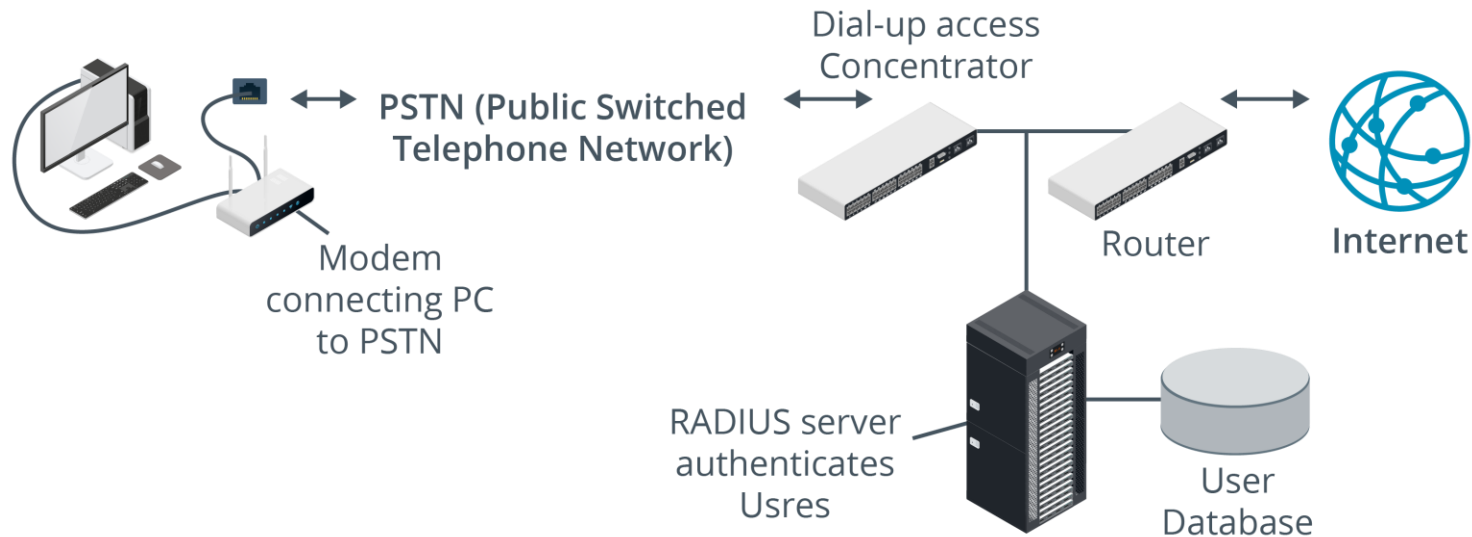
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Internet Service Providers (Slide 1 of 2)

- Home and business networks use ISP to connect to Internet.
 - Network connects to ISP's Point of Presence (PoP).
 - Dial-up, broadband (DSL, FTTx, cable), wireless connections.
 - Most use PSTN (aka POTS, "local loop," "last mile").
- ISP allocates IP addresses, registers domain names, hosts email and websites.
- Enterprise ISPs offer high bandwidth through fiber optic cable.

Internet Service Providers (Slide 2 of 2)



Broadband Internet Access

- A range of technologies
- “Always on”
- Data transfer rates much higher than dial-up

DSL (Slide 1 of 2)

- DSL uses high frequencies in digital phone line for communications.
- Filter separates DSL signals from voice traffic.
- Advanced modulation and echo cancelling enable high-bandwidth, full-duplex.
- DSL “modem” connects to phone system (usually router/modem/AP appliance).
- Phone line connects to DSL modem bank (DSLAM).
- PPP over ATM (PPPoA) or PPP over Ethernet (PPPoE).

DSL (Slide 2 of 2)

DSL Type	Description
Asymmetrical DSL (ADSL)	• Consumer version; fast downlink, slow uplink • Various iterations • ADSL2+: downlink rates up to ~24 Mbps; uplink rates up to ~1.4 Mbps • Providers may restrict data download • Cable quality, number of users may affect speed • Max range ~2 miles/3 km
Symmetric DSL	• Same uplink and downlink speeds • Useful for businesses, branch offices
Very High Bitrate DSL (VDSL)	• High bit rate at expense of range • Symmetric and asymmetric modes • Asymmetric: 52 Mbps downstream/6 Mbps upstream over 300 m/1000 ft • Symmetric: 26 Mbps in both directions • VDSL2: 100 Mbps bi-directional rate for very short range

Fiber Optic Internet Access (Slide 1 of 3)

- Higher bandwidth, longer distance than copper cable
- Has replaced copper as core of telecommunications networks
- Being extended to individual homes and businesses
- Two principal types of fiber optic network services:
 - Cable TV providers
 - Telecom providers

Fiber Optic Internet Access (Slide 2 of 3)

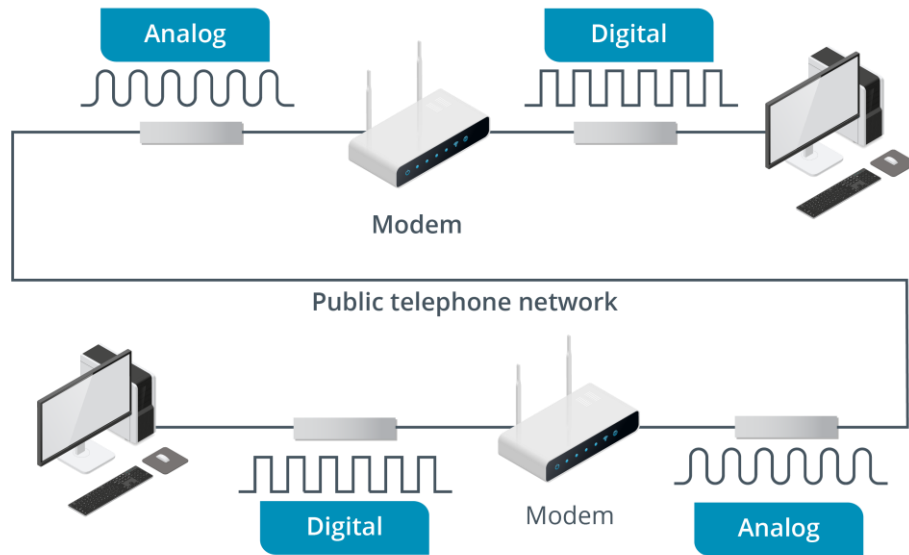
- Hybrid Fiber Coax (HFC)/Cable (“broadband cable” “cable”): Connection through CATV service, combines fiber core with coax to customer.
- Cable modem connects to local network through Ethernet adapter.
- Coax links all premises in a street with CMTS to ISP PoP via fiber backbone.
- DOCSIS: Downlink up to 38 Mbps (North America) or 50 Mbps (Europe); and uplink up to 27 Mbps.
- DOCSIS v3 allows multiplexed channels for higher bandwidth.

Fiber Optic Internet Access (Slide 3 of 3)

FTTx Solution	Description
Fiber to the X (FTTx)	• Solutions where fiber replaces copper in the “last mile”
Fiber to the Premises (FTTP) Fiber to the Home (FTTH)	• Most expensive, not widespread • Fiber link terminated at customer premises equipment
Fiber to the Node (FTTN) Fiber to the Curb/Cabinet (FTTC)	• Fiber to communications cabinet at street • Similar to HFC, but consumer link uses VDSL over phone wiring (not coax)

Dial-Up Internet Access (Slide 1 of 2)

- Telephone connection between computers.
- Uses entire frequency range; not efficient, low bandwidth.
- Phone charges apply; line cannot be used for voice at same time.
- Modems at each end convert digital $\leftrightarrow$ analog (MOfulation/DEModulation).



Dial-Up Internet Access (Slide 2 of 2)

- Disadvantages: low data transfer, time to establish connection, errors
- Fastest modems ~33.6 Kpbs; speed limit of phone line
- Theoretical maximum downlink ~56 Kpbs
- Compression may improve data transfer
- Has been superseded; still in use as a backup or for areas without other support

ISDN Internet Access

- Digital circuit-switched technology for voice, video, data.
- Uses copper telephone wiring if of sufficient quality.
- Uses digital signatures for both voice and data; no analog conversions.
- Dial-up service billed by line rental and usage; establishes connection in ~1 second.
- Used to connect LANs and for remote workers.
- Two classes:
 - BRI: two 64 Kbps "B" data channels for data and one 16 Kbps "D" control channel.
 - PRI: 23 or 30 "B" channels, one 24 Kbps "D" channel.
- Remains in use for telecom core; superseded by DSL/cable for subscribers.
- Terminal Adapter connects to PC or router; to network via NT1 device.

Fixed Wireless Internet Access (Slide 1 of 2)

- Wired broadband may not be available:
 - In rural areas
 - In older buildings where not possible to run new cable
- Fixed wireless may be an option
- Two options:
 - Satellite
 - Line of Sight (LoS) Wireless Internet Provider (WISP)

Fixed Wireless Internet Access (Slide 2 of 2)

Solution	Description
Satellite	• Large coverage area with VSAT microwave antenna aligned to orbital satellite. • Super High Frequency range (3-30 GHz). • Satellite television receivers for domestic use; use growing for businesses, especially rural. • Transfer rates vary: 6 Mbps / 15-20 Mbps down typical. • Can be severe latency problems. • Dish at customer aligned with satellite; connects via coax to DVB-S modem.
LoS WISP	• Ground-based microwave antennas aligned with each other; transmit if no physical obstruction (usually atop tall buildings). • Spans great distances; no cabling infrastructure; lower latency than satellite. • Hard to maintain line of sight; expensive. • WISP may use Wi-Fi or proprietary equipment. • Range of frequencies; may be affected by 5G cellular phone service deployment.

Cellular Radio Networks (Slide 1 of 2)

- Wi-Fi bands have restricted range; fixed wireless requires large antenna.
- Cellular radio wireless networking allows long-distance communications over smartphone devices.
- Also used by IoT devices.
- Connects to nearest transmitter; base station range of up to 5 miles.
- Transmitter connects phone to mobile/landline networks.
- 850 / 1900 MHz bands (Americas); 900 / 1800 MHz bands (rest of world).

Cellular Radio Networks (Slide 2 of 2)

Generation	Description
2G	• GSM phones using a SIM card; international, and AT&T in US • TIA/EIA IS-95 (cdmaOne) handsets managed by provider with CDMA; Sprint and Verizon • Data access built on top of existing voice network using CSD • Must establish data connection to base station, incurring charges; maximum ~14.4 Kpbs
3G	• Deployed packet-switched technology to mobiles • GPRS/EDGE; HSPA+; CDMA2000/Evolution Data Optimized (EV-DO)
4G	• LTE: converged 4G standard supported by all network providers, requires a SIM. Maximum 150 Mbps down; 20 Mbps real-world • LTE-A: Intended to provide 300 Mbps down; 40 Mbps current real-world
5G	• Target is for 1 Gbps if stationary or slow-moving; 100 Mbps if fast-moving • Available in trial areas; commercially in ~2020 • 70 Gpbs in test conditions (<i>← James is this right???? Should this be 70 Mbps? -LO</i>)

Activity



Discussing Internet Connection Types

Network Infrastructure Concepts

- Wired Networks
- Network Hardware Devices
- Wireless Networks
- Internet Connection Types
- Network Configuration Concepts
- Network Services

Routers (Slide 1 of 2)

- Switches use MAC addresses; routers use logical network and host IDs.
- Many different types and uses; two general tasks:
 - LAN router: divides a physical network into logical networks.
 - WAN (edge/border) router: joins separate networks (i.e.; LAN to Internet).
- Route/path to destination is selected either dynamically or statically; packet moves by hops along path to target.
- At target, hardware address determines destination node.



Routers (Slide 2 of 2)

- Routers and modems both connect to the Internet:
 - Modem makes a physical link (like a switch).
 - Router makes logical forwarding decisions.
 - Often bundled in one device.
- Switched enterprise networks can have thousands of ports; inefficient to treat as one logical network.
 - Use VLANs on managed switches to group ports into logical subnets.
 - VLANs communicate through routers.
 - Also provides filtering and monitoring to improve security.

The TCP/IP Protocol Suite (Slide 1 of 3)



Protocol: Rules and formats enabling systems to exchange data.

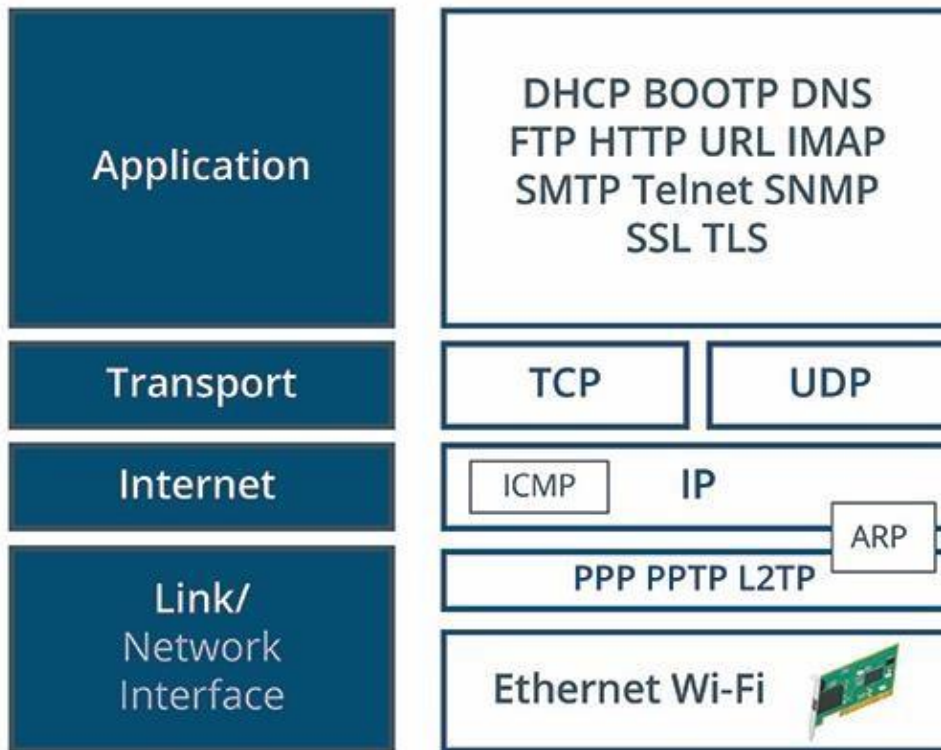
Protocol Suite: A collection of several protocols used for networking are designed to work together.

- Networks have converged on use of TCP/IP protocol suite
- Originally developed by US DoD; now an open standard
- IETF working groups implement development
- RFCs are published standards
- Packet-based protocols
- Routers select the path for packets
- Main protocols handle addressing and transport
- Divided into four-layer model

The TCP/IP Protocol Suite (Slide 2 of 3)

Layer	Description
Link/Network Interface Layer	• Puts frames on physical network • Not TCP/IP protocols as such; networking products and media (Ethernet, Wi-Fi) • Communications on local network • Data packaged in frames • Nodes identified by MAC address
Network Layer (IP Protocol)	• IP provides packet addressing and routing • Best-effort delivery; unreliable, connectionless
Transport Layer (TCP/UDP Protocols)	• TCP guarantees orderly packet transmission • UDP provides non-guaranteed packet transfer, but is faster
Application Layer	• Numerous protocols for network configuration, management, services; use TCP/UDP ports • ARP: Finds MAC address associated with IP address • ICMP: delivers status and error messages (used by <code>ping</code> and <code>tracert</code>)

The TCP/IP Protocol Suite (Slide 3 of 3)



Internet Protocol and IP Addressing (Slide 1 of 4)

- Two versions of IP, IPv4 and IPv6. Main headers in IPv4.

IPv4 Frame Field	Description
Source IP Address	Identifies the sender of the datagram by IP address.
Destination IP Address	Identifies the destination of the datagram by IP address.
Protocol	Indicates whether data should be passed to TCP or UDP at the destination.
Checksum	Verifies the packet's integrity at the destination.
Time to Live	• The number of hops the datagram can stay on the network before it is discarded; avoids endless looping of undeliverable packets. • Each router decreases the TTL value by at least one.

Internet Protocol and IP Addressing (Slide 2 of 4)

- IP address defines source and destination of packet:

32 binary digits:

11000110001010010001000000001001

Divided into octets:

11000110 00101001 00010000 00001001

Converted to dotted-decimal notation:

198 . 41 . 16 . 9

Internet Protocol and IP Addressing (Slide 3 of 4)

- In binary, a digit can only be 0 or 1
- Values of the digits are powers of 2
- Converting 11101101 from binary to decimal:

Place value:	128	64	32	16	8	4	2	1
Binary value:	1	1	1	0	1	1	0	1
Conversion:	$128*1$	$64*1$	$32*1$	$16*0$	$8*1$	$4*1$	$2*0$	$1*1$
Decimal equivalent:	$128 + 64 + 32 + 0 + 8 + 4 + 0 + 1 = 237$							

Internet Protocol and IP Addressing (Slide 4 of 4)

- Converting 199 from decimal to binary:

199 =

Decimal value:	128	+	64	+	0	+	0	+	0	+	4	+	2	+	1
Place value:	128		64		32		16		8		4		2		1
Conversion:	128*1		64*1		32*0		16*0		8*0		4*1		2*1		1*1
Binary equivalent:	1		1		0		0		0		1		1		1

- Maximum value of a byte is 255, minimum is 0.
- Theoretical address range is 0.0.0.0 to 255.255.255.255; some addresses not permitted or reserved.

Subnet Masks (Slide 1 of 4)

- IP address encodes both network ID and host ID.
- Subnet mask separates them by “masking” the host.
- Binary 1 in the mask = address digit is part of network ID.
- Size of the network portion of the subnet mask determines how many networks and hosts allowed in a given addressing scheme.
- Expressed in dotted decimal or as network prefix (contiguous number of 1s in the mask).

Subnet Masks (Slide 2 of 4)

- Default masks and network classes:

Class	Dotted Decimal Mask	Network Prefix	Binary Mask
A	255.0.0.0	/8	11111111 00000000 00000000 00000000
B	255.255.0.0	/16	11111111 11111111 00000000 00000000
C	255.255.255.0	/24	11111111 11111111 11111111 00000000

Subnet Masks (Slide 3 of 4)

- Network ID revealed by “ANDing”
- 1 AND 1 = 1; all other combinations = 0
- Example:

172.	30.	15.	12	10101100	00011110	00001111	00001100
255.	255.	0.	0	11111111	11111111	00000000	00000000
172.	30.	0.	0	10101100	00011110	00000000	00000000

Subnet Masks (Slide 4 of 4)

- Hosts communicate directly if on same network
- IP protocol uses subnet mask to compare source/destination network ID
- If on same network, delivers locally:

172.	30.	15.	12
255.	255.	0.	0.
172.	30.	16.	101

- If on different network, sends to router:

172.	30.	15.	12
255.	255.	0.	0.
172.	31.	16.	101

Host IP Configuration

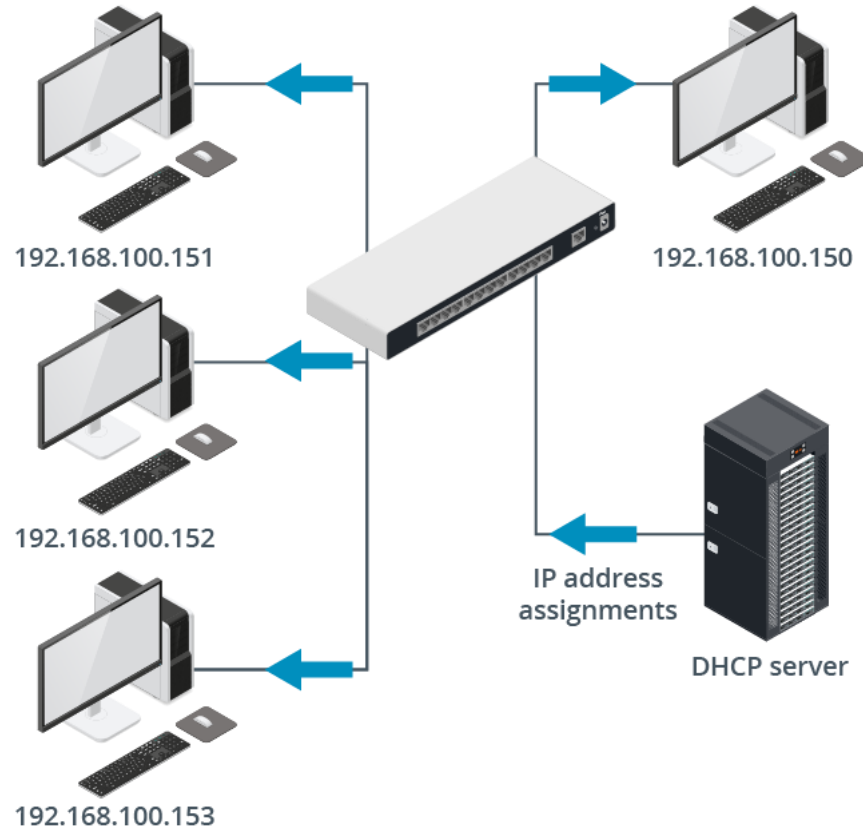
- Host must have IP address and subnet mask; should have other parameters for proper network/Internet communication.

Parameter	Description
IPv4 address Subnet mask	• Both required for every interface; can be set manually. • Address: dotted-decimal notation; identifies host and network. • Subnet mask determines if other hosts are local or remote.
Default gateway	• IP address of a router to send packets outside of local network. • If no gateway, host can only communicate on local network.
Client DNS	• IP address of DNS server to provide host/domain name resolution and locate Internet resources. • DNS also used on most local networks. • Often the gateway address; often a second server address provided for redundancy.

Static and Dynamic IP Addresses (Slide 1 of 3)

- Static addressing:
 - Administrator manually configures each host.
 - Must update manually if host changes subnet.
 - Must track address allocations to avoid duplication.
 - Can be time consuming and error-prone.
 - Only used for systems with dedicated functionality.
- Dynamic addressing:
 - DHCP server allocates addresses.

Static and Dynamic IP Addresses (Slide 2 of 3)



Static and Dynamic IP Addresses (Slide 3 of 3)

Dynamic Addressing Method	Description
DHCP	• DHCP client contacts server on boot and requests address. • Also provides other parameters (subnet mask, default gateway) . • Limited time leases. • Information configured on server; client updated when lease renewed.
Link local APIPA	• Fallback mechanism for DHCP client if DHCP server is unavailable. • Host self-configures with address on 169.254.x.x network. • “Link local” is generic term; “APIPA” is Microsoft term. • Communication with other APIPA hosts on same network only.
DHCP reservation	• For hosts that need same address each time. • Configure DHCP server with reserved host address per MAC address. • Centralized, easier to implement than static addressing.

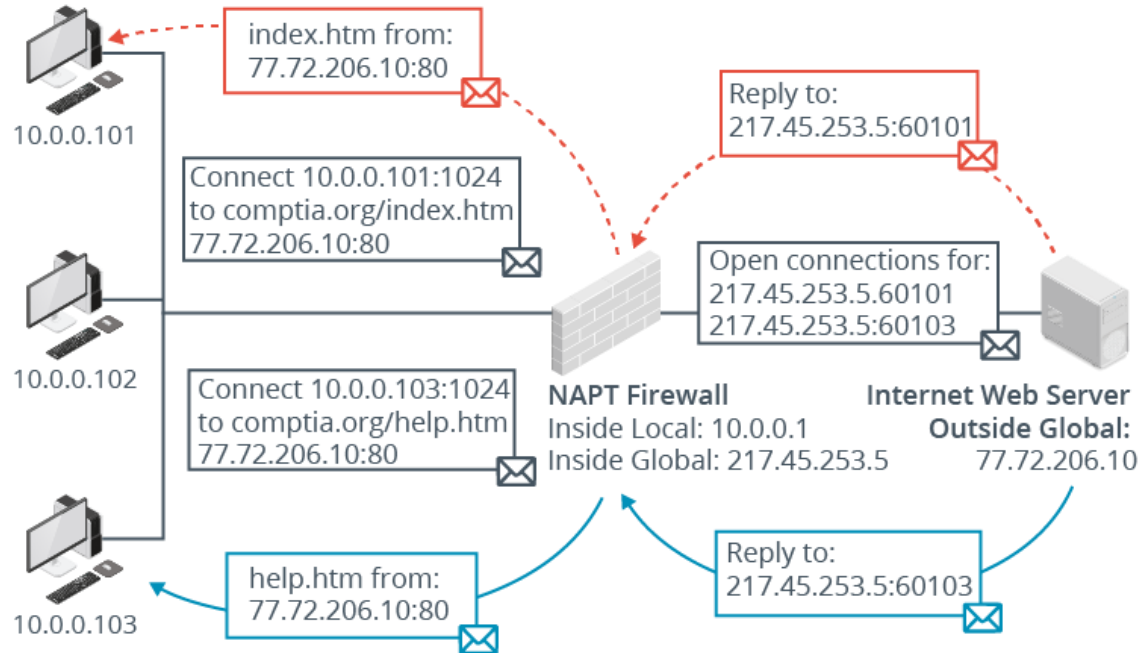
Public and Private IP Addresses (Slide 1 of 4)

- On the Internet, each host address must be unique; usually allocated by ISP.
- Few organizations have enough individual addresses; various methods to overcome this issue.
- Internal hosts can use addresses in a Class A, B, or C private range defined by RFC 1918 (10.0.0.0 to 10.255.255.255; 172.16.0.0 to 172.31.255.255; 192.168.0.0 to 192.168.255.255).
- Internet access provided for private-address hosts through:
 - A router using NAT.
 - A proxy server.

Public and Private IP Addresses (Slide 2 of 4)

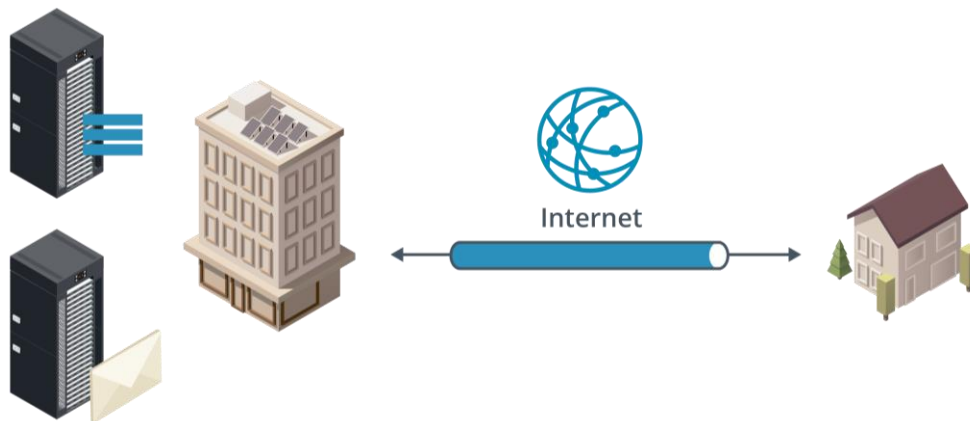
- In NAT, router converts the internal private IP address to a valid public address.
- IP configuration is simpler and internal clients are not directly accessible from the Internet.
- NAT address pool itself will be limited; multiple private addresses will use a single public address.
- Mapping provided by NAPT, aka PAT, aka NAT overloading.
 - Each outgoing connection assigned TCP or UDP port.
 - Returning traffic mapped back to address/client port.

Public and Private IP Addresses (Slide 3 of 4)



Public and Private IP Addresses (Slide 4 of 4)

- VPN: connects two private networks over a public network (the Internet).
- Internet is cost-effective way to connect users and networks, but is not private.
- VPN protocols create tunnels through the public network to authenticate, encrypt, and secure private communications.



IPv6 (Slide 1 of 5)

- IPv4 address pool is large, but limited.
- IPv6 uses 128-bit addresses, massively increasing address pool.
- Other improvements: simplified address headers, hierarchical addressing, support for time-sensitive traffic, new unicast address structure.
- Large string of characters in binary or even decimal; affects clarity and accuracy.
- Uses hexadecimal notation (0-9, A-F):

Decimal	Hex	Binary
0	0	0000
1	1	0001
...	...	...
10	A	1010
11	B	1011
...	...	...

IPv6 (Slide 2 of 5)

- Binary IPv6 address divided into eight double-byte values using hex notation:

2001:0db8:0000:0000:0abc:0000:def0:1234

- Leading zeros can be ignored, and a contiguous series of zeroes can be replaced by a double colon place marker:

2001:db8::abc:0:def0:1234

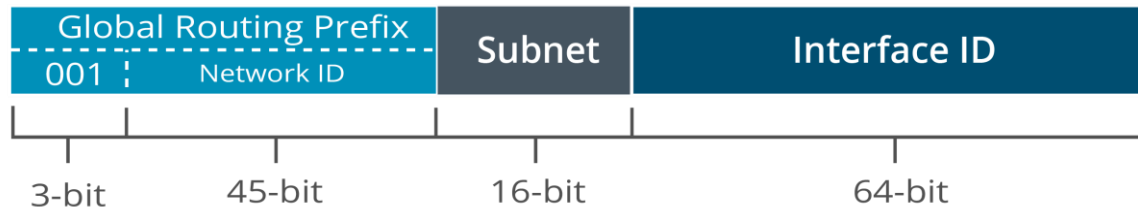
IPv6 (Slide 3 of 5)

- First 64 bits are network ID, second 64 bits designate the interface
- Fixed size = no subnet mask; /nn = length of routing prefix in bits



IPv6 (Slide 4 of 5)

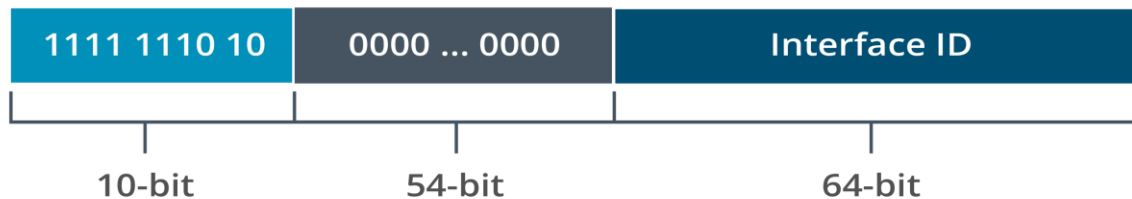
- IPv6 global unicast address format:



- IPv6 address blocks assigned hierarchically by routers; logical address space

IPv6 (Slide 5 of 5)

- IPv6 link-local addresses used for housekeeping
- Span single subnet
- Nodes on same link called “neighbors”
- Start with fe80::
- Equivalent of APIPA
- IPv6 host always has a link-local address



Activity



Discussing Network Configuration Concepts

Activity



Network Infrastructure Concepts

- Wired Networks
- Network Hardware Devices
- Wireless Networks
- Internet Connection Types
- Network Configuration Concepts
- Network Services

TCP and UDP Ports (Slide 1 of 2)

- Transport-layer protocols ensure effective delivery; content of packets is significant.
- Identifies network application types by assigning port number (0-65535).
- Data from upper layers is packaged in segments, tagged with port number.
- Passed to network layer for delivery.
- Simultaneous segment transmissions are multiplexed onto network link; de-multiplexed at receiving host.
- Can use TCP or UDP.

TCP and UDP Ports (Slide 2 of 2)

Port Type	Description
TCP	• Ensures reliability and sequencing with acknowledgement messages. • If non-delivery, retransmits if lack of acknowledgement. • If damaged delivery, NACK forces retransmission. • Connection-oriented. • Acknowledgements add overhead, slow communications.
UDP	• Connectionless, non-guaranteed, no sequencing or flow control. • Speeds up communication by reducing overhead. • For applications that: • Don't require acknowledgement and can tolerate missing or out-of-order packets. • Are time-sensitive but don't need complete reliability.

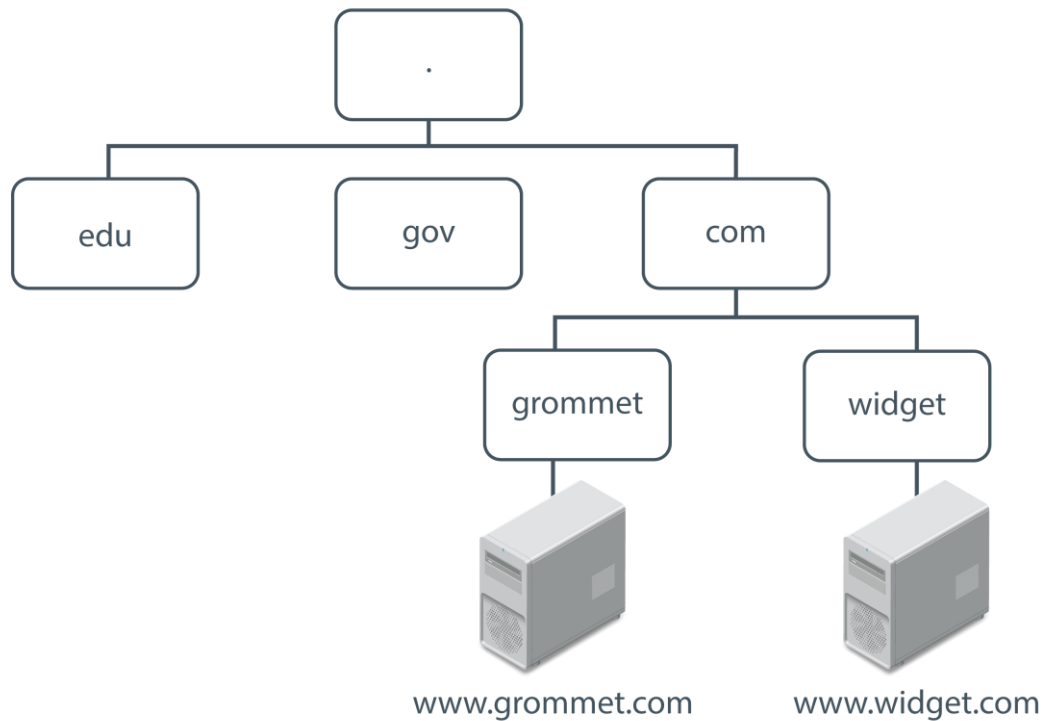
Well-Known Ports

- Port: unique ID for a service using TCP or UDP for transport.
- Might be persistent (for servers) or ephemeral (for clients).
- IANA assigns standard (“well-known”) port numbers to services.
 - See course text for examples.
- IANA defines ephemeral port range (49152 to 65535); some OSes use different values.
- Firewalls must have ports enabled or disabled to allow only valid traffic.

DNS (Slide 1 of 4)

- Hierarchical system for resolving names to IP addresses.
- Database distributed among many name servers; distributes maintenance, protects against server loss.
- Root (.) at top; then 13 TLDs (generic, sponsored, or country code); then domains.
- Domain names managed by ICANN, registered with the appropriate Domain Name Registry for the TLD.
- Records traced from root down; each level of server has information about servers below in hierarchy.

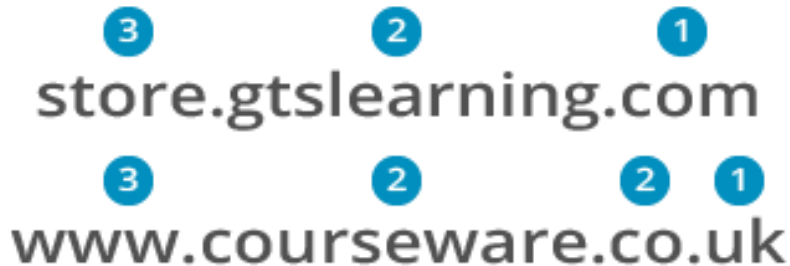
DNS (Slide 2 of 4)



DNS (Slide 3 of 4)

- FQDN shows hierarchy from most specific on left to least specific on right.
- Domain name portion identifies the company, organization, or individual; must be unique and officially registered.
- Host name identifies particular server or server alias.

store.gtslearning.com
www.courseware.co.uk



The diagram illustrates the hierarchy of two domain names. For 'store.gtslearning.com', the labels are 3 above 'store', 2 above 'gtslearning', and 1 above 'com'. For 'www.courseware.co.uk', the labels are 3 above 'www', 2 above 'courseware', 2 above 'co', and 1 above 'uk'. This demonstrates how the hierarchy moves from most specific on the left to least specific on the right.

DNS (Slide 4 of 4)

DNS Server Type	Description
Authoritative name server	• Holds domain records and can respond authoritatively about hosts in the domains it manages. • Required for Active Directory. • If private domain, not available outside the LAN; on Internet, published to name servers hosted by ISPs.
Recursive resolver	• Resolves names for clients. • Client contacts resolver; resolver contacts name servers until record is located or request times out. • DNS clients are configured with resolver address. • Listens on UDP 53.

Web Servers and HTTP/HTTPS (Slide 1 of 3)



Web server: A server that provides client access using HTTP (defaults to port 80) or its secure version HTTPS (defaults to port 443).

- Organizations may lease from ISP; host directly; use private servers (intranets)
- Provides HTML pages (text files with tags), interpreted by browsers
- Extended by scripts and web applications

Web Servers and HTTP/HTTPS (Slide 2 of 3)

- Uses URL to access resources:
 1. Protocol
 2. FQDN
 3. File path


http://store.gtslearning.com/comptia/index.htm

Web Servers and HTTP/HTTPS (Slide 3 of 3)

- HTTP lacks security; data sent unencrypted, no authentication.
- SSL/TLS can be used to encrypt TCP/IP applications that use TCP connections, including HTTPS.
- Servers use digital certificates from Certification Authorities to prove the identity of the server and to provide encryption.



Mail Servers (Slide 1 of 3)

- Email can send text and file attachments encoded using MIME.
- Can use multiple protocols; typical process:
 1. Client sends message to server; server queues message for an SMTP session (port 25).
 2. SMTP server uses DNS to resolve address of recipient's mail server.
 3. SMTP delivers message; usually several "hops."
 4. Message placed in store on recipient's server; client software connects with mailbox using POP3 (port 110) or IMAP (port 143).
 - POP3 more widely used; IMAP has more features.
- Email account requires username, password, email address, incoming and outgoing server addresses, and protocol types.

Mail Servers (Slide 2 of 3)

Add Account

POP and IMAP Account Settings

Enter the mail server settings for your account.

User Information

Your Name:

David Martin

Email Address:

david@davidmartin.me

Server Information

Account Type:

POP3

Incoming mail server:

pop3.mysisp.net

Outgoing mail server (SMTP):

smtp.mysisp.net

Logon Information

User Name:

david

Password:

☒ Remember password

☒ Require logon using Secure Password Authentication (SPA)

Test Account Settings

We recommend that you test your account to ensure that the entries are correct.

Test Account Settings ...

☒ Automatically test account settings when Next is clicked

Deliver new messages to:

☒ New Outlook Data File

☐ Existing Outlook Data File

Browse

More Settings ...

< Back

Next >

Cancel

Mail Servers (Slide 3 of 3)

- Mailto URL scheme: username@domainname (domain may be a company or ISP).
- Different systems allow different characters; not usually treated as case sensitive.
- Mail may be rejected if incorrectly addressed, if identified as spam, if mailbox is full.
- Only one of many network communication types.



File and Printer Sharing (Slide 1 of 3)

- Core network functions.
- May be accomplished by proprietary protocols (i.e., File and Print Services for Windows).
- May use standard protocols (i.e., FTP), but may not have as much functionality.

File and Printer Sharing (Slide 2 of 3)

Protocol	Description
SMB (aka CIFS)	• Underpins file and printer sharing on Windows networks; currently SMB2, but legacy clients are supported. • TCP port 445; also NetBIOS over TCP/IP (UDP and TCP port range 137-139). • Implemented as Samba on Linux.
AFP	• Performs similar function to SMB for Apple/Mac OS. • UDP or TCP port 427 (Service Location Protocol)—not required by OS X or later. • TCP port 548.
FTP	• Early TCP/IP protocol; widely used for file transfers; flexible; easy to maintain. • TCP port 21 for connection; port 20 for active transfer or server-assigned port if passive. • Client options: • Command line • Dedicated GUI • Browsers

File and Printer Sharing (Slide 3 of 3)

Site Options -

Host Info

Startup

Transfer

Advanced

SSL

Host Info

Site Name: gtslearning

Host name: www.gtslearning.com

User ID: admin

Password: xxxxxxxx ☒ Save password

Account: ☐ Save account

Comment:

OK Cancel Help

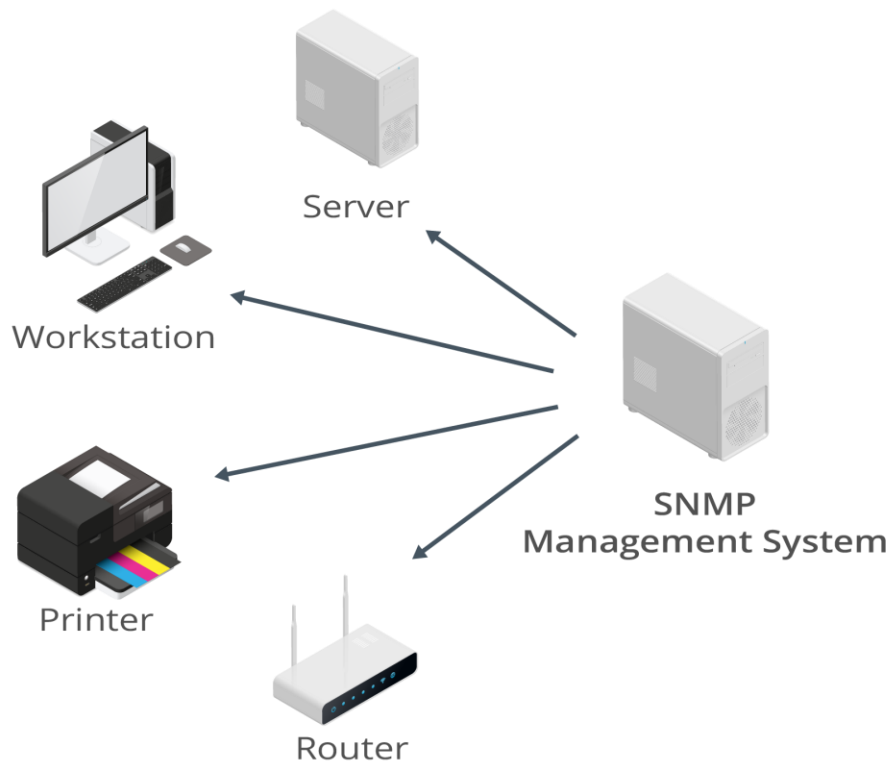
Network Host Services

Service	Description
Authentication Server	• Used on enterprise networks to ensure only authorized users can access accounts. • On Windows domain, Active Directory provides authentication based on Kerberos. • AAA server consolidates authentication across multiple devices. • RADIUS is an AAA protocol.
DHCP DNS	• DHCP assigns IP addresses to hosts when they connect. • DNS allows hosts to access resources by host name and FQDN by resolving names to IP.
LDAP	• Network resources are recorded as objects in a directory database. • X.500 standards allow directories to interact; full standard required a complex protocol. • LDAP allows X.500-compliant queries and updates over TCP/IP. • Widely supported; TCP/UDP 389. • Uses Distinguished Names and Relative Distinguished Names as identifiers.
NetBIOS/NetBT	• NetBIOS first Windows network software; provided name discovery, addressing. • NetBT runs NetBIOS over TCP and UDP ports 137-139 (name services, datagram transmission, session services). • Should be disabled unless supporting legacy Windows systems or appliances.

Inventory Management Servers (Slide 1 of 2)

Service	Description
SNMP	• Framework for managing/monitoring network devices. • Management system and agents. • Agent process runs on network device; maintains MIB; can initiate trap for a notable event. • System software provides oversight location, monitors agents, displays information. • Device queries=UDP 161; traps=UDP 162.
Endpoint Management	• Facilitates Defense in Depth security policies that require hardening to workstation level. • Can apply OS and anti-virus updates; catalog software; apply security policies; analyze logs; monitor performance and alerts. • Example: Microsoft's SCCM.
syslog	• Helpful to consolidate separate device logs. • Prior to Windows 7, Windows logs were local; 3rd-party tools used to consolidate. • Windows event subscription can forward log events to central system. • UNIX and Linux equivalent is syslog. • Client-server model for event collection; open format; <i>de facto</i> standard.

Inventory Management Servers (Slide 2 of 2)



Legacy and Embedded Systems

System Type	Description
Embedded	• Designed for a specific function. • Range from individual microcontrollers to complex industrial control systems. • May have been designed for a closed network, without connectivity. • Special design and security considerations when interacting with a data network. • Risk for maintenance and troubleshooting; require specialist knowledge.
Legacy	• No longer supported by vendor. • May be retained on networks to support existing services that are not practical to migrate. • Security risks. • Should be isolated from network. • Like embedded systems, risk for maintenance and troubleshooting; require specialist knowledge.

Internet Security Appliances and Software

System Type	Description
IDS/NIDS	• Software and/or hardware that monitors for and quickly detects malicious behavior. • Can also analyze and alert administrators to infrastructure problems. • Can comprise sensors, detection software, and management software; each implementation is unique.
IPS/NIDS	• Inline security device that monitors for and blocks suspicious network and system traffic. • May drop packets, reset connections, sound alerts; at times quarantine intruders. • Examines packet contents. • UTM appliance combines firewall, A-V scanner, and IDS.
Proxy Server	• Used on enterprise networks as alternative to NAT. • Checks and forwards HTTP, email, or other requests from internal hosts to Internet; returns reply to the client. • May be transparent (no client configuration) or non-transparent (client must be configured with proxy's IP address and port, typically 8080).

Activity



Discussing Network Services

Reflective Questions

1. What do you think are the most important network concepts covered in this lesson?
2. What experience do you have with any of the technologies discussed in this lesson?

